MATH3603

UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS
Probability Theory
Test 1

1. Let the probability density function of X be given by

$$f(x) = \begin{cases} c(1-x^2), & -1 < x < 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) What is the value of c?

Solution.

$$\int_{-\infty}^{\infty} f(x)dx = \int_{-1}^{1} c(1-x^2)dx = c(2 - \frac{2}{3}) = \frac{4c}{3}.$$

Therefore, if f is a density function, we must have $c = 3/4$. □

(b) What is the cumulative distibution function of X?

Solution.

$$F(x) = \int_{-\infty}^{x} f(y)dy = \begin{cases} 0, & x \le -1 \\ \int_{-1}^{x} \frac{3(1-y^2)}{4}dy, & -1 < x < 1 \\ 1, & x \ge 1 \end{cases} = \begin{cases} 0, & x \le -1 \\ \frac{3x-x^3+2}{4}, & -1 < x < 1 \\ 1, & x \ge 1 \end{cases}$$

□

2. Let X_i, $i = 1, 2, \ldots,$ be independent random variables, each being uniformly distributed over $(0, 1)$. Estimate $\mathbb{P}(\sum_{i=1}^{10} X_i > 7)$.

Solution.

$$\mu = \mathbb{E}[X_i] = \int_0^1 xdx = \frac{1}{2}, \;\; \sigma^2 = \mathrm{Var}(X_i) = \int_0^1 (x - \frac{1}{2})^2 dx = \frac{1}{12}.$$

We have by the central limit theorem:

$$\begin{aligned} \mathbb{P}(\sum_{i=1}^{10} X_i > 7) &= \mathbb{P}\left(\frac{\sum_{i=1}^{10} X_i - 10\mu}{\sigma\sqrt{10}} > \frac{7 - 10\mu}{\sigma\sqrt{10}} \right) \simeq 1 - F_Z(\frac{7-5}{\sqrt{10/12}}) \\ &\simeq 1 - F_Z(2.19) \\ &= 1 - 0.98574 = 0.01426. \end{aligned}$$

□

3. Suppose that there are w white balls and r red balls in a box where w and r are positive integers. Let P_n be the probability of an even number of white balls being drawn with replacement at the nth draw and we define $P_0 = 1$.

(a) Derive a recurrence relation between P_{n+1} and P_n for $n = 0, 1, 2 \ldots,$.

Solution. We have

$$P_{n+1} = \frac{r}{w+r}P_n + \frac{w}{w+r}(1 - P_n) = \frac{w}{w+r} + (1 - \frac{2w}{w+r})P_n.$$

Because there are two cases at the nth draw: even (probability is P_n) and odd (probability is $1 - P_n$). If it is even, then we must have a red ball at the $(n+1)$th draw and if it is odd then we must have a white ball at the $(n+1)$th draw. □

(b) Hence find P_{n+1} and $\lim_{n\to\infty} P_{n+1}$.

Solution. Let $\alpha = w/(w+r)$ where $0 < \alpha < 1$ then we have

$$P_{n+1} = \alpha + (1 - 2\alpha)P_n = \alpha + \alpha(1 - 2\alpha) + (1 - 2\alpha)^2 P_{n-1}.$$

Inductively we have

$$P_{n+1} = \alpha(1 + (1 - 2\alpha) + (1 - 2\alpha)^2 + \cdots + (1 - 2\alpha)^n) + (1 - 2\alpha)^{n+1} P_0.$$

Hence

$$P_{n+1} = \frac{\alpha(1 - (1 - 2\alpha)^{n+1})}{2\alpha} + (1 - 2\alpha)^{n+1} = \frac{1}{2} + \frac{1}{2}(1 - 2\alpha)^{n+1}.$$

Since $|1 - 2\alpha| < 1$, we have

$$\lim_{n\to\infty} P_{n+1} = 0.5.$$

□

4. Suppose Y is uniformly distributed on $(0, 1)$, and that the conditional distribution of X given that $Y = y$ is uniform on $(0, y)$. Find

(a) $\mathbb{E}[X]$;

Solution. We have:

$$f_{X|Y}(x|y) = \begin{cases} \frac{1}{y}, & 0 < x < y \\ 0, & \text{otherwise} \end{cases}$$

Thus,

$$\mathbb{E}[X|Y = y] = \int_0^y \frac{1}{y} x dx = \frac{y}{2}.$$

And

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X|Y]] = \int_0^1 \frac{y}{2} dy = \frac{1}{4}.$$

□

(b) $\mathrm{Var}(X)$.

Solution.

$$\mathbb{E}[(X-\mathbb{E}[X])^2|Y=y] = \int_0^y \frac{1}{y}\left(x-\frac{1}{4}\right)^2 dx = \frac{y^2}{3}-\frac{y}{4}+\frac{1}{16}.$$

Thus,

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[\mathbb{E}[(X-\mathbb{E}[X])^2|Y]] \\ &= \int_0^1 \frac{y^2}{3}-\frac{y}{4}+\frac{1}{16}dy \\ &= \frac{y^3}{9}-\frac{y^2}{8}+\frac{y}{16}\Big|_0^1 = \frac{1}{9}-\frac{1}{16} = \frac{7}{144}.\end{aligned}$$

□

5. An individual traveling on the real line is trying to reach the origin. However, the larger the desired step, the greater is the variance in the result of that step. Specifically, whenever the person is at location x, he next moves to a location having mean 0 and variance βx^2. Let X_n denote the position of the individual after having taken n steps. Supposing that $X_0 = x_0$, find

(a) $\mathbb{E}[X_n\,|X_{n-1}]$ and $\mathrm{Var}[X_n\,|X_{n-1}]$;

Solution.

$$\mathbb{E}[X_n\,|X_{n-1}] = 0,\ \ \mathrm{Var}[X_n\,|X_{n-1}] = \beta X_{n-1}^2.$$

□

(b) $\mathbb{E}[X_n]$;

Solution.

$$\mathbb{E}[X_n] = \mathbb{E}[\mathbb{E}[X_n\,|X_{n-1}]] = 0.$$

□

(c) $\mathrm{Var}(X_n)$.

Solution.

$$\mathrm{Var}(X_n) = \mathbb{E}[(X_n-\mathbb{E}[X_n])^2] = \mathbb{E}[X_n^2].$$

Now,

$$\begin{aligned}\mathbb{E}[X_n^2] &= \mathbb{E}[\mathbb{E}[X_n^2|X_{n-1}]] \\ &= \mathbb{E}[\mathbb{E}[(X_n-\mathbb{E}[X_n])^2|X_{n-1}]] \\ &= \mathbb{E}[\mathrm{Var}[X_n|X_{n-1}]] \\ &= \beta\mathbb{E}[X_{n-1}^2] = \cdots = \beta^n\mathbb{E}[X_0^2] = \beta^n x_0^2.\end{aligned}$$

□

6. Data indicate that the number of traffic accidents in Berkeley on a rainy day is a Poisson random variable with mean 9, whereas on a dry day it is a Poisson random variable with mean 3. Let X denote the number of traffic accidents tomorrow. If it will rain tomorrow with probability 0.6, find

   (a) $\mathbb{E}[X]$;

   Solution. Let $Y = 1$ if it rains tomorrow and $Y = 0$ otherwise. We have

   $$\mathbb{E}[X|Y = 1] = 9, \ \mathbb{E}[X|Y = 0] = 3.$$

   Therefore,

   $$\mathbb{E}[X] = \mathbb{E}[X|Y = 1]\mathbb{P}(Y = 1) + \mathbb{E}[X|Y = 0]\mathbb{P}(Y = 0) = 9 * 0.6 + 3 * 0.4 = 6.6.$$

   □

   (b) $\mathbb{P}(X = 0)$;

   Solution. Recall that a Poisson random variable with parameter λ has mean λ. Therefore

   $$\mathbb{P}(X = 0|Y = 1) = e^{-9}, \ \mathbb{P}(X = 0|Y = 0) = e^{-3}.$$

   Therefore,

   $$\mathbb{P}(X = 0) = \mathbb{P}(X = 0|Y = 1)\mathbb{P}(Y = 1)+\mathbb{P}(X = 0|Y = 0)\mathbb{P}(Y = 0) = 0.6e^{-9}+0.4e^{-3}.$$

   □

   (c) $\mathrm{Var}(X)$.

   Solution. Since a Poisson random variable with parameter λ has variance λ, we have:

   $$\mathbb{E}[X^2|Y = 1] = 9^2 + 9 = 90, \ \mathbb{E}[X^2|Y = 0] = 3^2 + 3 = 12.$$

   Thus,

   $$\mathbb{E}[X^2] = \mathbb{E}[X^2|Y = 1]\mathbb{P}(Y = 1)+\mathbb{E}[X^2|Y = 0]\mathbb{P}(Y = 0) = 90*0.6+12*0.4 = 58.8.$$

   Finally,

   $$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 58.8 - (6.6)^2 = 15.24.$$

   □